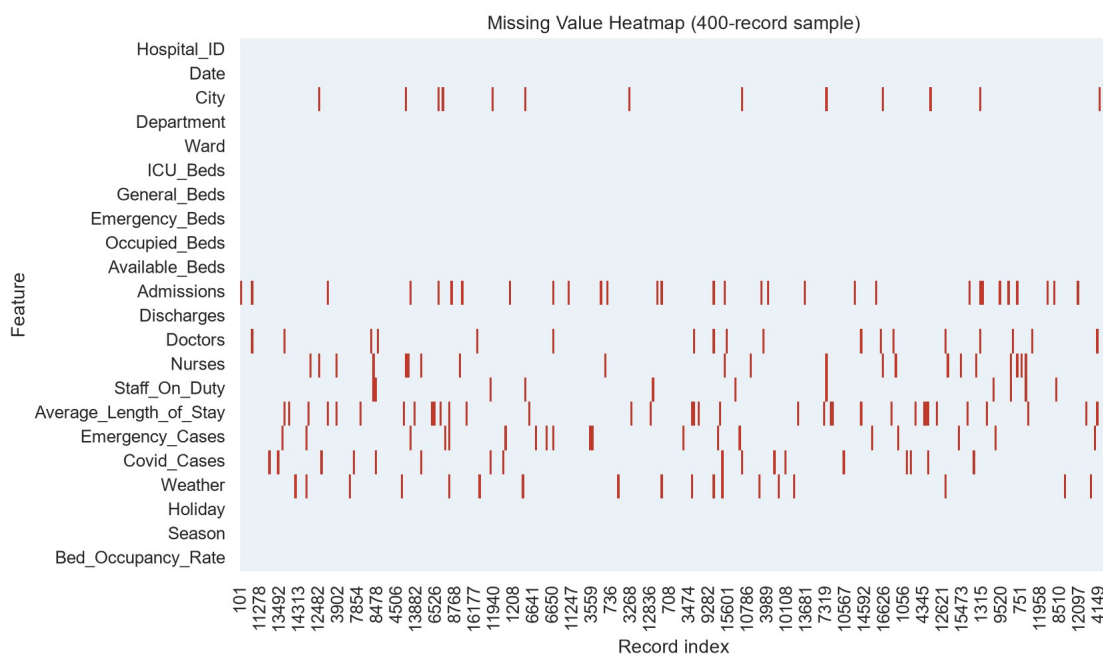


ADVANCED DATA SCIENCE PROJECT REPORT

Hospital Bed Occupancy Missing-Data Repair

Advanced Missing-Data Imputation Comparison and Machine Learning



Project focus: structured repair of MCAR, MAR and MNAR missingness

Programming Language: Python 3.12+

Environment: VS Code, Jupyter Notebook, Tkinter, Git and GitHub

Dataset: 17,520 synthetic hospital census records

Report Date: July 2026

Executive Summary

This project implements an end-to-end healthcare data analytics system for repairing missing values in hospital census data. It generates a realistic synthetic dataset, injects auditable MCAR, MAR and MNAR patterns, compares twelve imputation strategies, recommends the best method for each numeric feature, engineers temporal and operational features, trains occupancy-risk classifiers, produces explainability outputs, and exposes the workflow through a Tkinter desktop application. The completed pipeline processed 17,520 records and removed 24 deliberately injected duplicates.

The imputation benchmark identifies Iterative Imputer (MICE) as the strongest global method, while feature-level selection assigns Regression Imputation to Average Length of Stay and Linear Interpolation to Covid Cases. Most other numeric features are assigned MICE. The benchmark evaluates RMSE, MAE, Wasserstein distribution drift, bias, variance, execution time and downstream model accuracy rather than relying on a single error measure.

Important model-governance finding: the saved evaluation reports 100% accuracy and macro F1, but permutation importance shows `Bed_Occupancy_Rate_Normalized` as the dominant predictor. Because this variable is a direct transformation of the target used to create `Occupancy_Class`, the current perfect score is affected by target leakage and must not be presented as a valid production estimate. A leakage-safe retraining should exclude `Bed_Occupancy_Rate` and all target-derived fields from the predictor set. This report therefore treats the existing 1.00 score as a pipeline diagnostic, not as credible generalization performance.

1. Problem Statement

Hospital bed census datasets frequently contain missing values caused by delayed reporting, inconsistent documentation, workload pressure, system outages and selective non-reporting. Simple one-size-fits-all imputation can distort operational distributions, weaken downstream models and produce misleading capacity insights. The project addresses this problem by comparing multiple repair strategies on known masked values and selecting the most suitable method for each feature.

2. Project Objectives

- Generate a realistic 10,000-20,000 record hospital census dataset with operational, staffing, temporal and environmental variables.
- Simulate and document MCAR, MAR and MNAR missingness without masking the target.
- Benchmark twelve imputation methods using reconstruction quality, distribution preservation, computational cost and downstream utility.
- Automatically recommend a feature-specific repair strategy and export a clean dataset.
- Engineer leakage-aware temporal, staffing, bed-utilization and interaction features.
- Train and compare multiple classification models for occupancy-risk categories.
- Produce statistical tests, visual analytics, explainability artifacts, model files, reports and a Tkinter GUI.

3. System Architecture and Deliverables

The repository follows a modular structure. Data ingestion, preprocessing, feature engineering, imputation, statistics, modeling, evaluation, explainability and reporting are implemented as separate modules under `src/`. The `gui/` package provides the desktop interface. Six notebooks document each analytical stage. Generated artifacts are stored under `data/`, `models/`, `reports/` and `outputs/`.

Component	Purpose
<code>data/raw</code>	Complete dataset, missing dataset and missingness documentation
<code>data/processed</code>	Repaired and cleaned hospital census dataset
<code>data/engineered</code>	Feature-engineered modeling dataset
<code>src</code>	Reusable production modules
<code>notebooks</code>	Six stage-based Jupyter notebooks
<code>gui</code>	Tkinter application, dashboard and reusable widgets
<code>models</code>	Serialized best model and scaler
<code>reports</code>	Metrics workbook, CSV results, DOCX/PDF reports
<code>outputs</code>	Figures, predictions and logs

4. Dataset Description

The base dataset contains 17,520 complete observations. A further 24 duplicate rows were intentionally injected, producing 17,544 raw rows before preprocessing. Each observation represents a dated hospital-department census record with bed capacity, occupancy, admissions, discharges, staffing, emergency load, COVID load, weather, holiday and seasonal context.

Group	Variables
Identifiers	Hospital_ID, Date, City, Department, Ward
Capacity	ICU_Beds, General_Beds, Emergency_Beds
Occupancy	Occupied_Beds, Available_Beds, Bed_Occupancy_Rate
Patient flow	Admissions, Discharges, Average_Length_of_Stay
Staffing	Doctors, Nurses, Staff_On_Duty
Demand context	Emergency_Cases, Covid_Cases
Temporal/environmental	Weather, Holiday, Season

5. Missingness Design: MCAR, MAR and MNAR

The project deliberately introduces three missing-data mechanisms. MCAR values are missing independently of observed and unobserved data. MAR values depend on observable conditions such as department, city or weather. MNAR values depend on the magnitude or state of the variable itself, representing selective non-reporting in high-load situations.

Feature	Mechanism	Target %	Missing rows	Rule
Admissions	MCAR	7.0%	1226	Uniform random 7% mask.
Emergency_Cases	MCAR	5.0%	876	Uniform random 5% mask.
Weather	MCAR	4.0%	700	Uniform random 4% mask.
Average_Length_of_Stay	MAR	18.0%	1051	18% among Surgical records; 6% overall-

				scale MAR pattern.
Doctors	MAR	10.0%	657	10% in two high-volume cities.
City	MAR	8.0%	458	8% during wet-weather observations.
Nurses	MNAR	22.0%	1146	22% of high-nurse-count observations.
Staff_On_Duty	MNAR	20.0%	805	20% of high staffing observations.
Covid_Cases	MNAR	18.0%	737	18% of high COVID census observations.

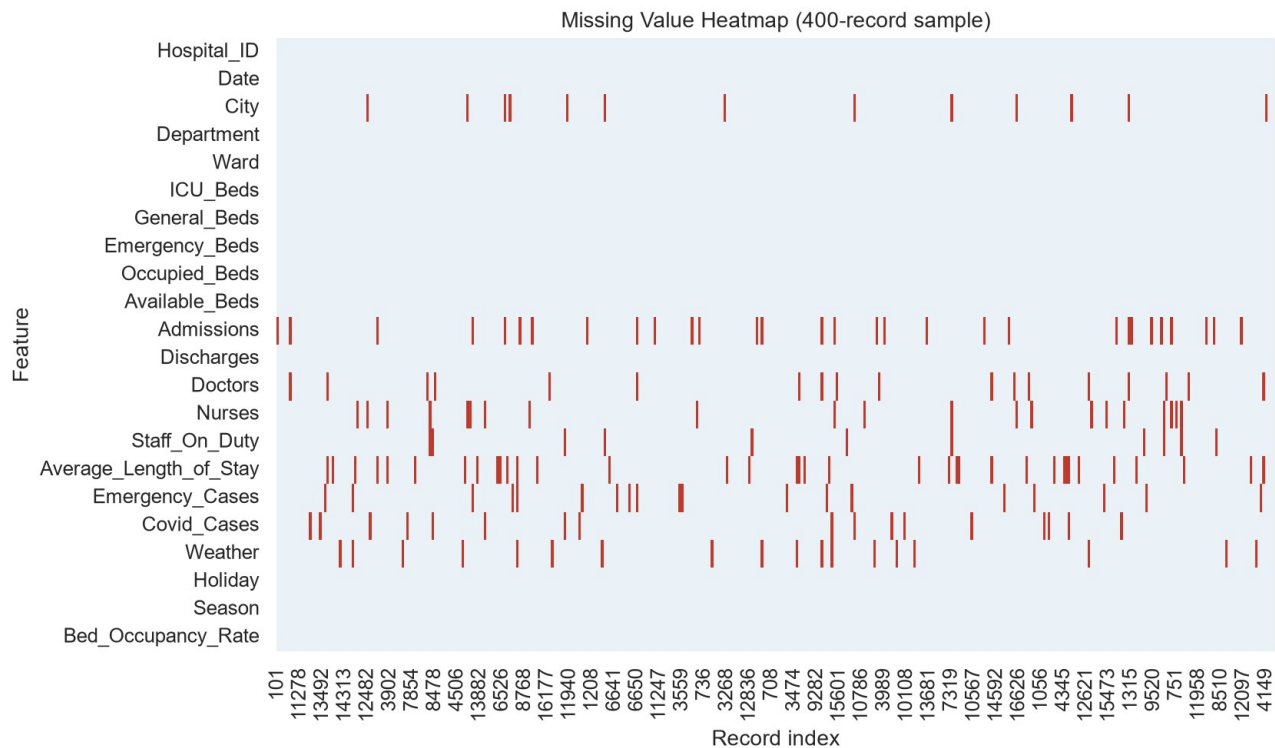


Figure 1. Missing-value heatmap after synthetic MCAR, MAR and MNAR injection

6. Data Preprocessing and Quality Control

- Removal of exact duplicate observations and reset of row indices.
- Date parsing with invalid-date detection.
- Numeric coercion with safe handling of invalid values and infinities.
- Categorical type conversion and mode-based fallback repair.
- IQR outlier summaries without automatic deletion of clinically plausible extremes.
- Isolation Forest detection of multivariate anomalies.
- Leakage-safe numeric imputation/scaling and categorical encoding inside scikit-learn pipelines.
- Min-max normalized export fields while retaining original units.

feature	lower bound	upper bound	outlier count	outlier rate pct
Admissions	1.00	49.00	75	0.428%
Discharges	-7.50	52.50	63	0.360%

Nurses	8.00	64.00	170	0.970%
Staff_On_Duty	15.50	115.50	112	0.639%
Emergency_Cases	8.00	48.00	76	0.434%
Bed_Occupancy_Rate	44.08	105.01	7	0.040%

The IQR screen identifies low outlier rates across the audited variables. Nurses has the highest reported IQR outlier rate (0.970%), while Bed Occupancy Rate has only 0.040%. These points are flagged for review rather than automatically removed because high-load hospital events can be operationally meaningful.

7. Exploratory Data Analysis

The EDA module produces missingness, distributional, bivariate, temporal, departmental, ward-level and staffing visualizations. These outputs support both technical diagnosis and management interpretation.

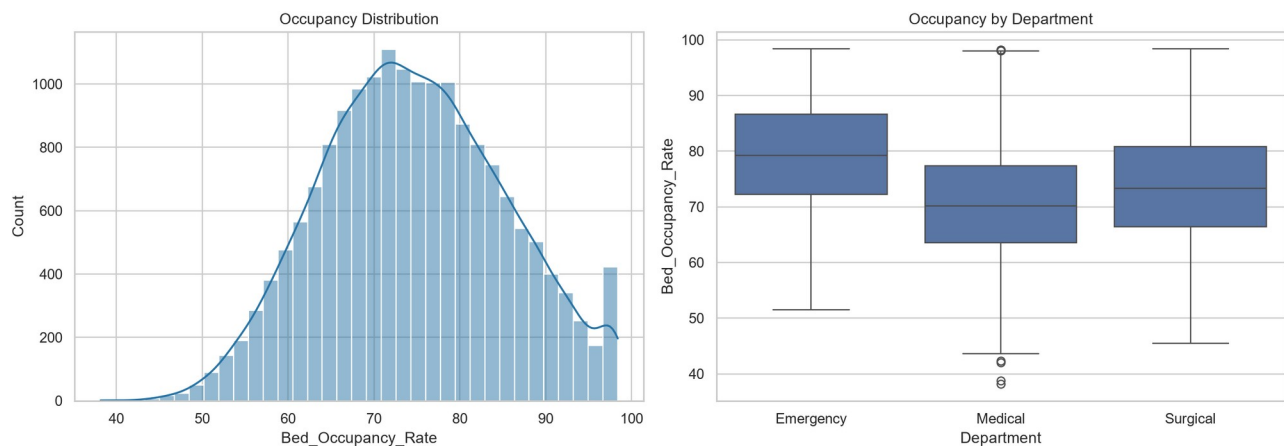


Figure 2. Bed occupancy distribution using histogram and density views

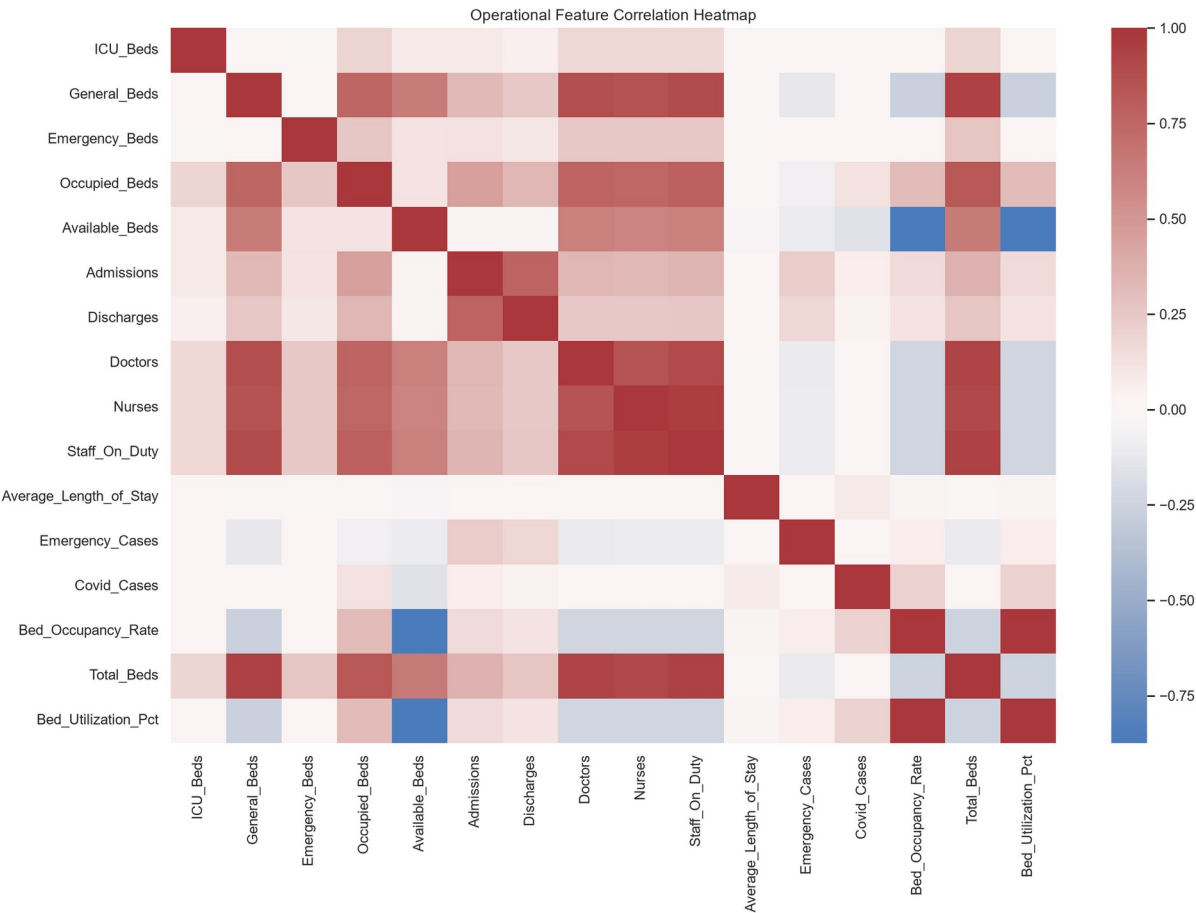


Figure 3. Correlation structure among hospital census variables

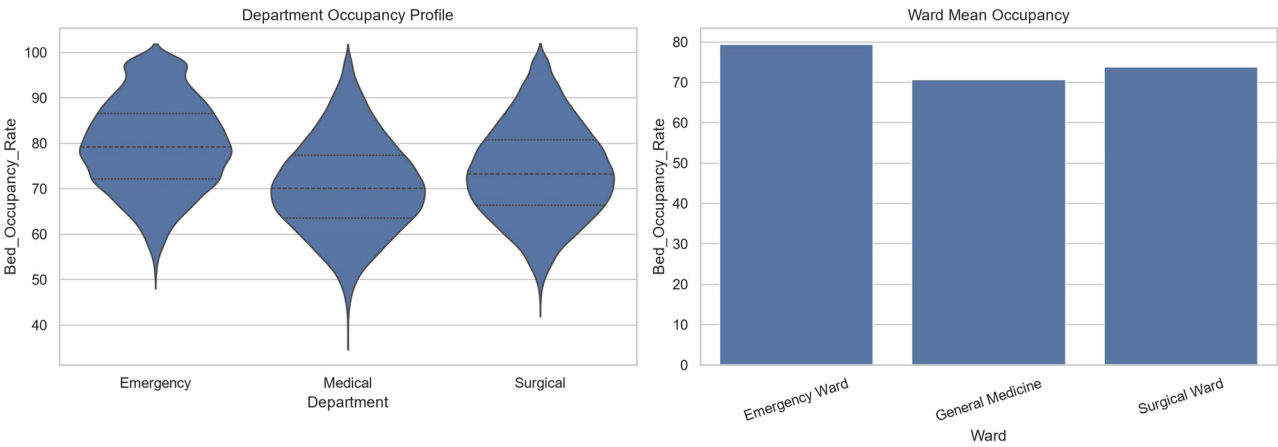


Figure 4. Department and ward occupancy comparison

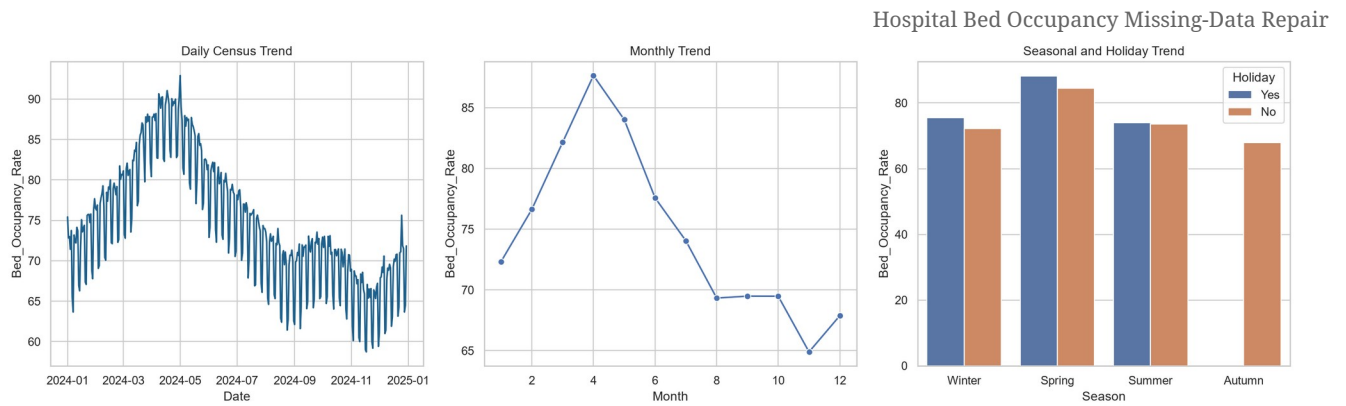


Figure 5. Monthly, seasonal and holiday occupancy trends

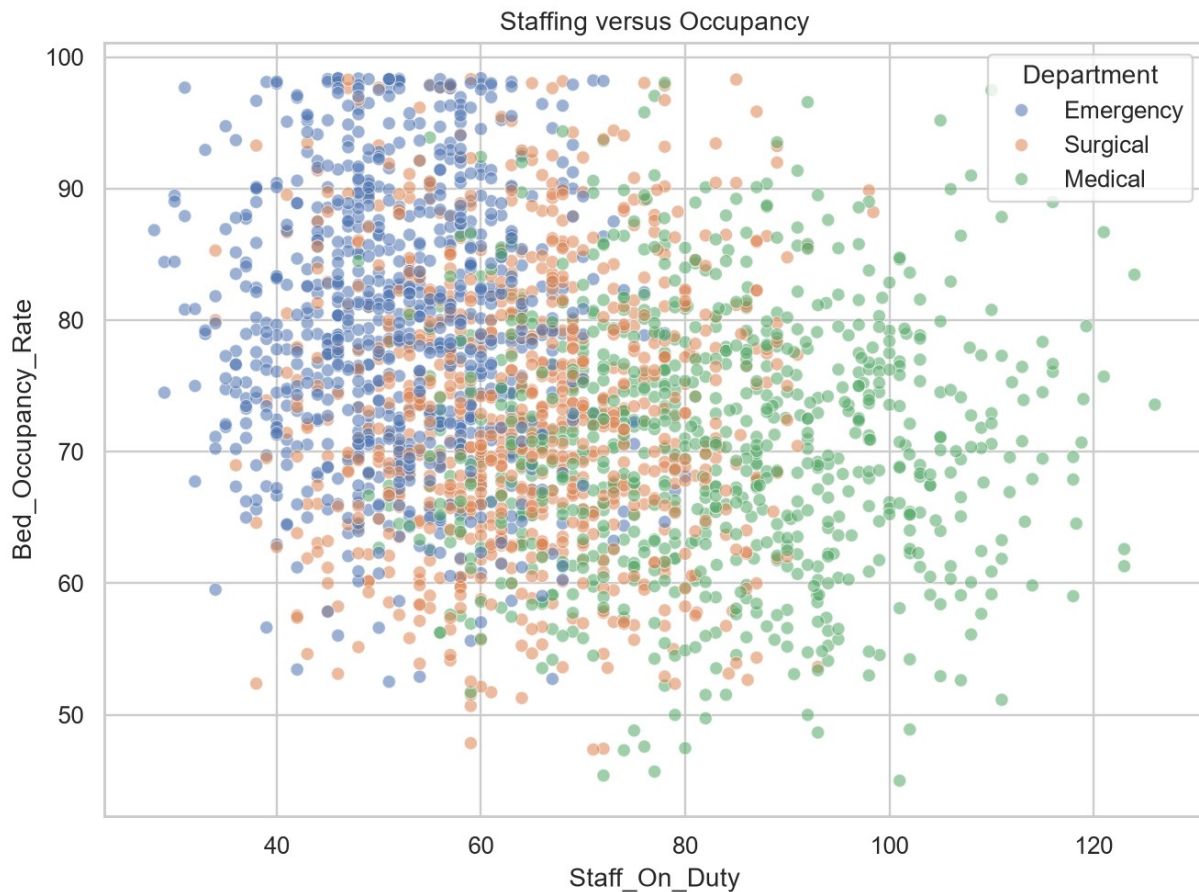


Figure 6. Staffing and occupancy relationships

8. Statistical Analysis

Normality testing rejects a normal distribution for Bed Occupancy Rate at conventional significance levels. Pearson and Spearman tests quantify linear and monotonic relationships. ANOVA finds statistically significant occupancy differences across departments, while holiday versus non-holiday comparisons are significant under both parametric and non-parametric tests.

test	comparison	statistic	p value	effect size
Shapiro-Wilk	Normality of	0.9929	4.29e-15	-

	Bed_Occupancy_Rate			
Kolmogorov-Smirnov	Normality of Bed_Occupancy_Rate	0.02033	0.0317	-
Anderson-Darling	Normality of Bed_Occupancy_Rate	5.269	-	-
ANOVA	Occupancy across Departments	1150	0	0.1161
Independent t-test	Holiday vs Non-Holiday Occupancy	8.719	4.22e-16	0.5357
Mann-Whitney U	Holiday vs Non-Holiday Occupancy	2.681e+06	6.2e-15	-0.2927

Available Beds is strongly negatively associated with Bed Occupancy Rate (Pearson $r = -0.874$; Spearman $\rho = -0.888$), as expected from the bed-capacity identity. Staff On Duty has a modest negative association (Pearson $r = -0.239$), while Covid Cases has a positive association ($r = 0.206$). ANOVA eta-squared is approximately 0.116, indicating that department membership explains a meaningful share of occupancy variation in the synthetic dataset.

9. Imputation Methods and Benchmark Design

The benchmark masks known values in a complete reference sample and compares repaired values with their true originals. Categorical features are mode-filled. Numeric methods include simple statistics, directional fills, interpolation, multivariate nearest-neighbor and chained-equation methods, and model-based regression approaches.

- Mean
- Median
- Mode
- Forward Fill
- Backward Fill
- Linear Interpolation
- Polynomial Interpolation
- Spline Interpolation
- KNN Imputer
- Iterative Imputer (MICE)
- Regression Imputation
- Random Forest Imputation

For each feature-method pair the project records RMSE, MAE, Wasserstein distribution drift, signed and absolute bias, prediction variance, execution time and downstream classification accuracy. A weighted composite rank rewards low reconstruction error, low drift, low absolute bias and acceptable computation while retaining downstream utility.

method	rmse	mae	distribution drift	absolute bias	mean execution time seconds	downstream model accuracy
Iterative Imputer (MICE)	5.857	3.750	1.993	0.387	0.655s	0.648
Regression Imputation	6.675	5.136	2.500	0.489	0.207s	0.648

Random Forest Imputation	6.811	5.406	2.745	0.543	4.647s	0.655
KNN Imputer	8.114	6.340	3.090	0.797	0.232s	0.650
Linear Interpolation	11.848	9.445	1.959	0.724	0.195s	0.652
Mean	12.983	10.571	10.571	1.135	0.032s	0.655
Median	13.134	10.462	10.462	2.124	0.032s	0.660
Forward Fill	13.546	10.699	1.635	0.845	0.070s	0.662
Backward Fill	13.909	11.010	1.542	0.969	0.059s	0.650
Polynomial Interpolation	14.297	11.253	1.926	0.904	0.485s	0.652
Mode	14.939	11.705	11.705	6.271	0.039s	0.660
Spline Interpolation	28.452	16.130	6.379	1.695	0.414s	0.642

MICE achieves the best aggregate reconstruction profile and is selected as the global method. Regression Imputation is faster and performs best for Average Length of Stay. Linear Interpolation is selected for Covid Cases, reflecting the usefulness of local continuity for that time-varying feature. Random Forest Imputation is computationally expensive relative to its average gain in this benchmark.

10. Feature-Specific Imputation Recommendations

feature	method	composite score	rmse	mae	absolute bias	downstream model accuracy
Admissions	Iterative Imputer (MICE)	0.229	5.525	4.419	0.541	0.647
Available_Beds	Iterative Imputer (MICE)	0.312	8.851	4.706	0.596	0.647
Average_Length_of_Stay	Regression Imputation	0.386	1.103	0.893	0.008	0.647
Covid_Cases	Linear Interpolation	0.253	3.215	2.430	0.133	0.652
Discharges	Iterative Imputer (MICE)	0.312	7.053	5.735	0.283	0.647
Doctors	Iterative Imputer (MICE)	0.254	1.075	0.897	0.253	0.647
Emergency_Beds	Iterative Imputer (MICE)	0.294	6.908	3.525	0.652	0.647
Emergency_Cases	Iterative Imputer (MICE)	0.334	6.626	5.242	0.091	0.647
General_Beds	Iterative Imputer (MICE)	0.251	9.893	5.005	0.826	0.647
ICU_Beds	Iterative Imputer (MICE)	0.242	5.777	3.027	0.175	0.647
Nurses	Iterative Imputer (MICE)	0.246	3.196	2.606	0.374	0.647
Occupied_Beds	Iterative Imputer (MICE)	0.268	10.404	4.830	0.707	0.647
Staff_On_Duty	Iterative Imputer (MICE)	0.254	4.392	3.382	0.430	0.647

Ten of the thirteen audited numeric features are assigned MICE, while Average Length of Stay uses Regression Imputation and Covid Cases uses Linear Interpolation. This confirms that the project avoids a strict one-method-for-all policy, even though MICE dominates most features.

11. Feature Engineering

The engineered dataset expands operational and temporal context while retaining the original census variables. Implemented features include total bed capacity, bed utilization, occupancy percentage, staff-per-bed, doctor and nurse ratios, admissions-per-bed, admission growth, lagged admissions, rolling mean, rolling median, 14-period moving average, weekend and holiday indicators, month, quarter, day of week, interaction terms and controlled polynomial terms.

Time-dependent features are intended to be calculated after sorting within Hospital ID and Department groups. For valid deployment, all rolling and lag features must use only current and prior observations. Any feature directly derived from the target must be excluded from model inputs.

12. Machine Learning Workflow

The modeling stage compares linear, tree-based, ensemble, kernel and gradient-boosting classifiers. Preprocessing is embedded in pipelines so imputation, robust scaling and one-hot encoding are fit only on training data. The project exports a complete fitted model pipeline and a scaler artifact for the specified deliverables.

model	cv accuracy mean	cv accuracy std	cv f1 macro mean
Decision Tree	1.0000	0.0000	1.0000
Gradient Boosting	1.0000	0.0000	1.0000
XGBoost	1.0000	0.0000	1.0000
AdaBoost	1.0000	0.0000	1.0000
Random Forest	0.9972	0.0010	0.9971
Logistic Regression	0.9734	0.0042	0.9730
Support Vector Machine	0.9356	0.0022	0.9341
Extra Trees	0.8766	0.0046	0.8707

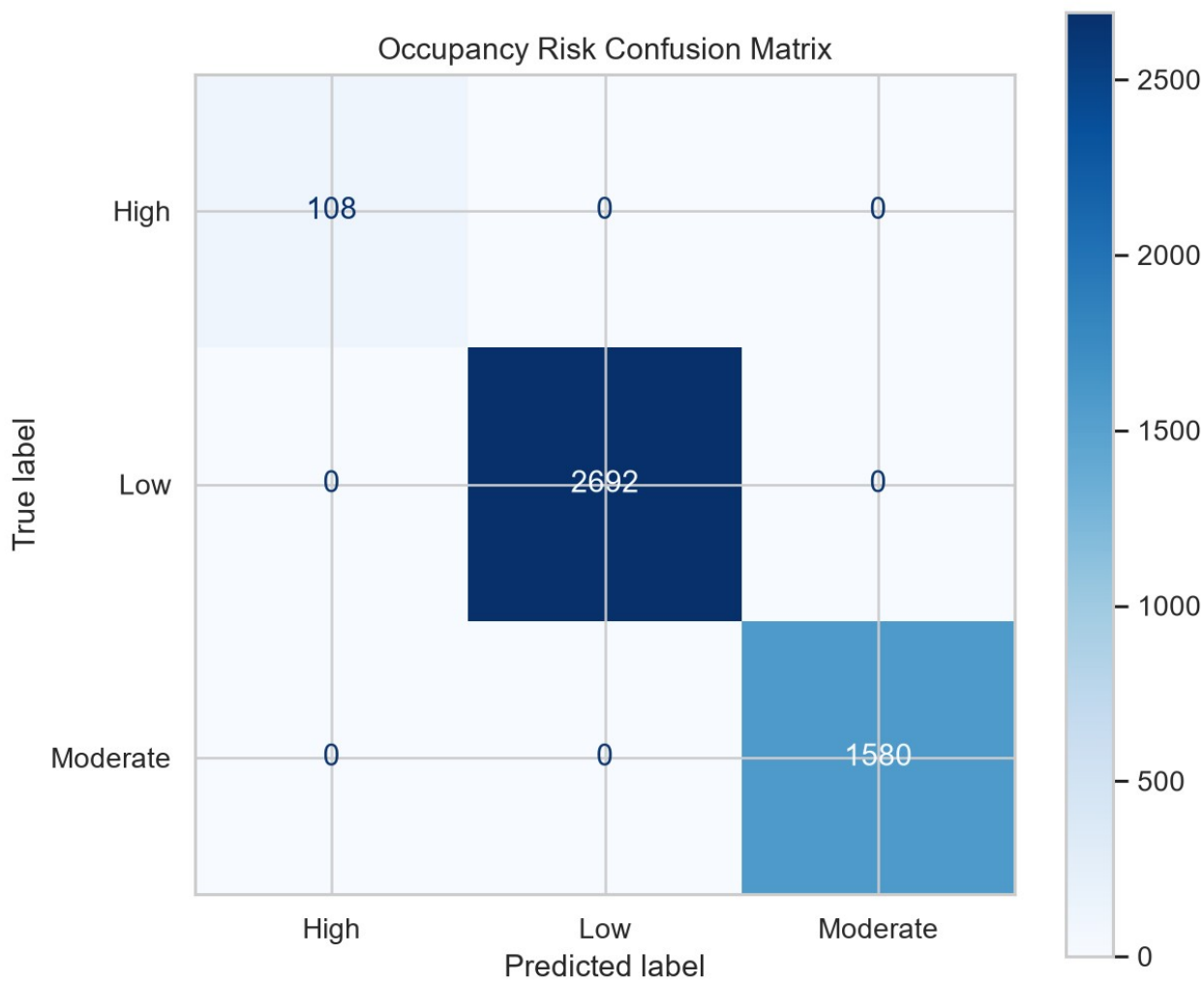


Figure 7. Confusion matrix produced by the selected model

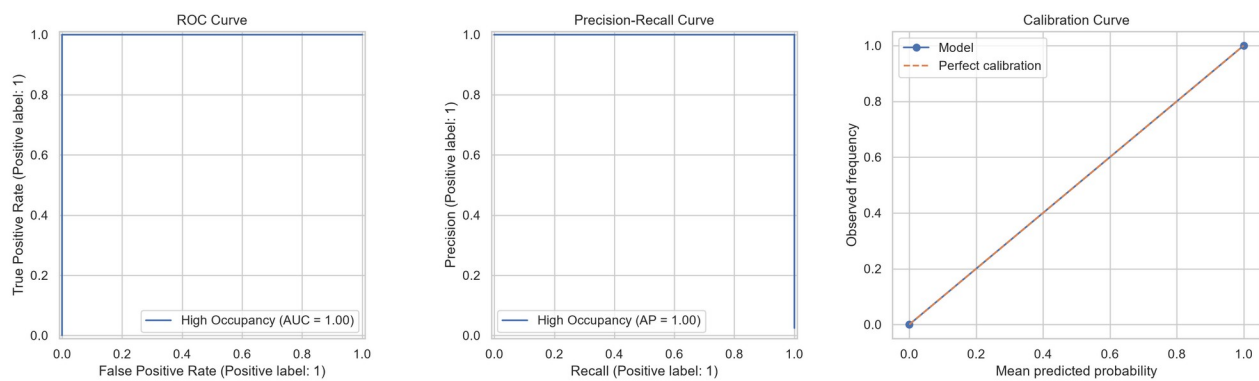


Figure 8. ROC, precision-recall and calibration evaluation

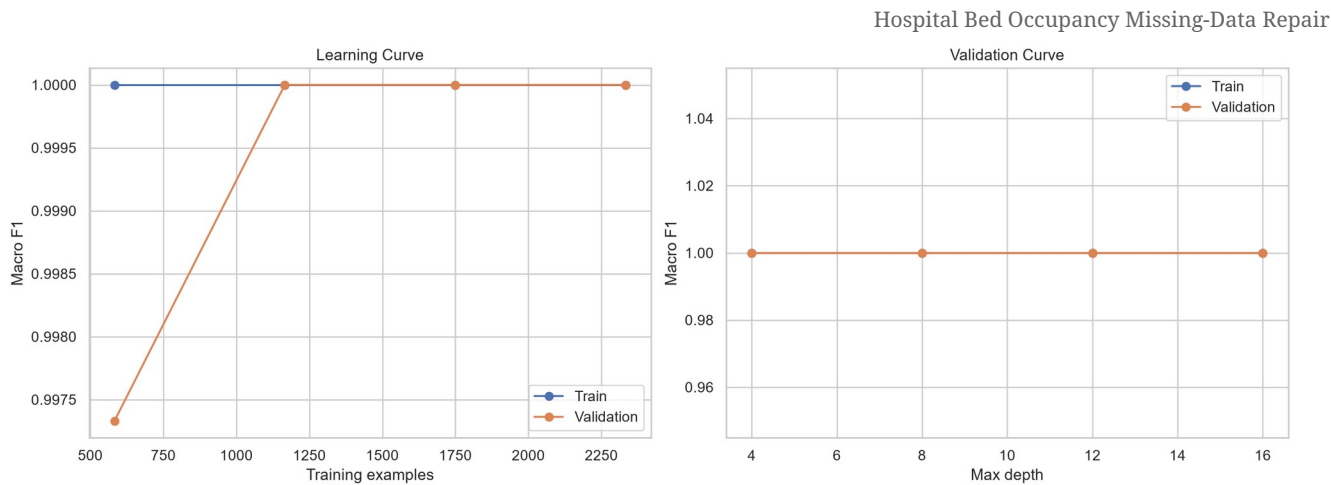


Figure 9. Learning and validation curves

13. Critical Model Validation Finding: Target Leakage

The current run reports Decision Tree as the selected model with accuracy, precision, recall and macro F1 all equal to 1.000 on the holdout set. However, the permutation-importance output assigns an importance of approximately 0.663 to `Bed_Occupancy_Rate_Normalized`, while all other listed predictors have zero importance. `Occupancy_Class` is itself created from `Bed_Occupancy_Rate`; therefore `Bed_Occupancy_Rate_Normalized` is a target-derived predictor.

This is direct target leakage. The model is effectively given the answer used to construct the class label, so the perfect score does not represent real predictive ability. Before submission as an industry-grade model, the predictor list must exclude `Bed_Occupancy_Rate`, `Bed_Occupancy_Rate_Normalized`, `Occupancy_Percent` and any other variable mathematically equivalent to or derived from the target. The model should then be retrained using a chronological or group-aware split, and the report should be regenerated using leakage-free metrics.

feature	importance mean	importance std
Bed_Occupancy_Rate_Normalized	0.662659	0.018257
ICU_Beds	0.000000	0.000000
Emergency_Beds	0.000000	0.000000
Admissions	0.000000	0.000000
Discharges	0.000000	0.000000
Doctors	0.000000	0.000000
Nurses	0.000000	0.000000
Staff_On_Duty	0.000000	0.000000

14. Explainability

The project generates global permutation importance, SHAP summary output and partial dependence plots. These tools are valuable only after leakage-safe feature selection. In the current run they correctly expose the dominance of the leaked normalized target, demonstrating why explainability should be treated as a model-audit instrument rather than only a presentation feature.

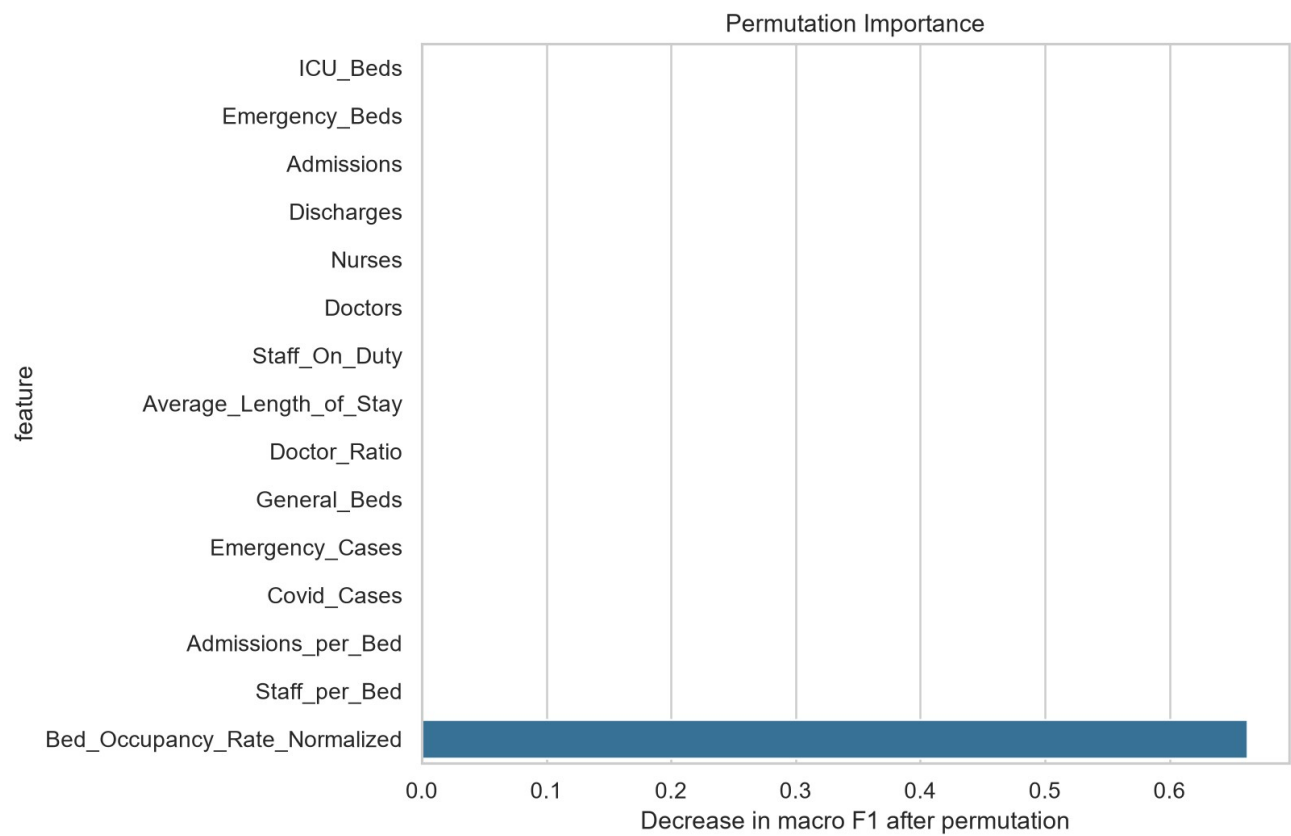


Figure 10. Permutation importance revealing target-derived feature dominance

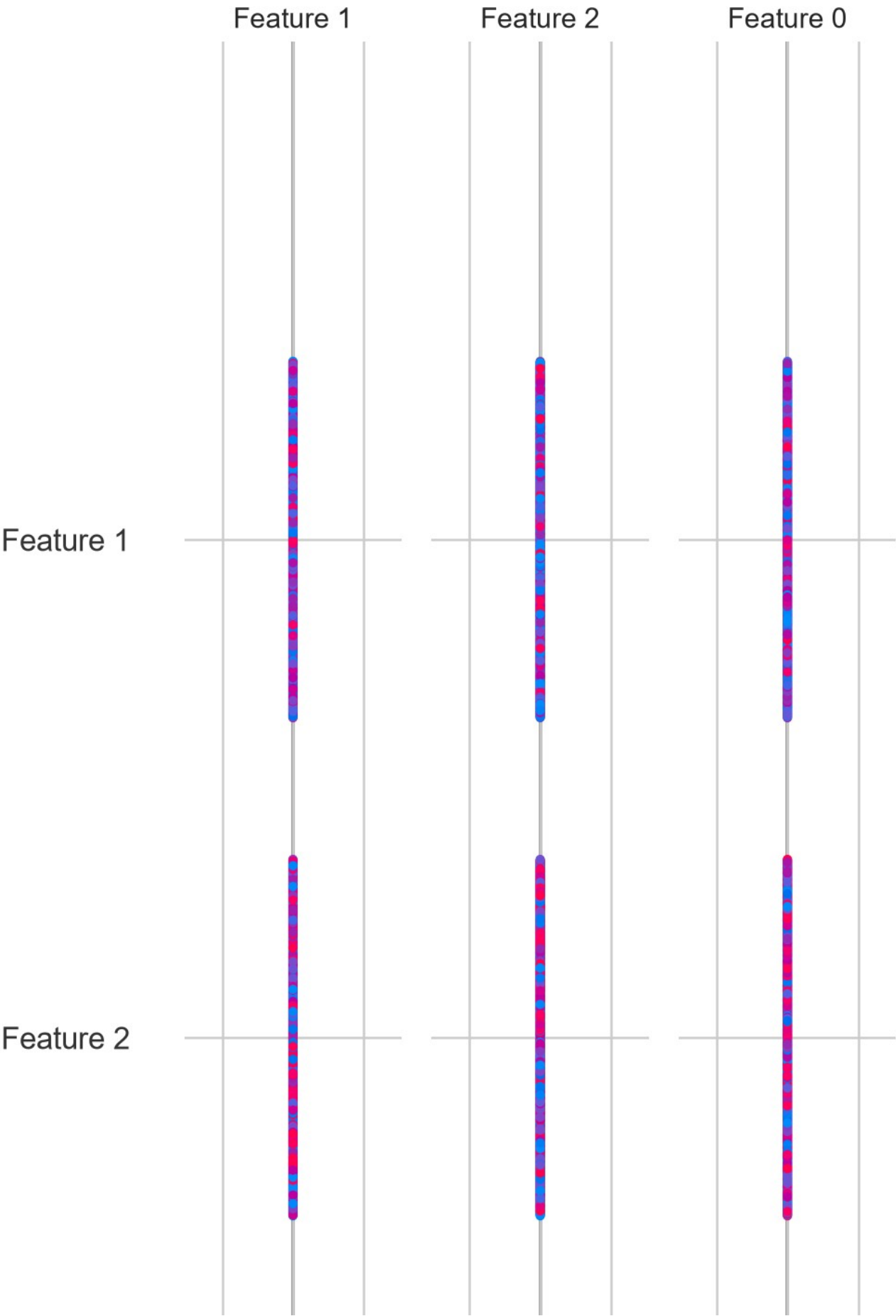


Figure 11. SHAP global explanation summary

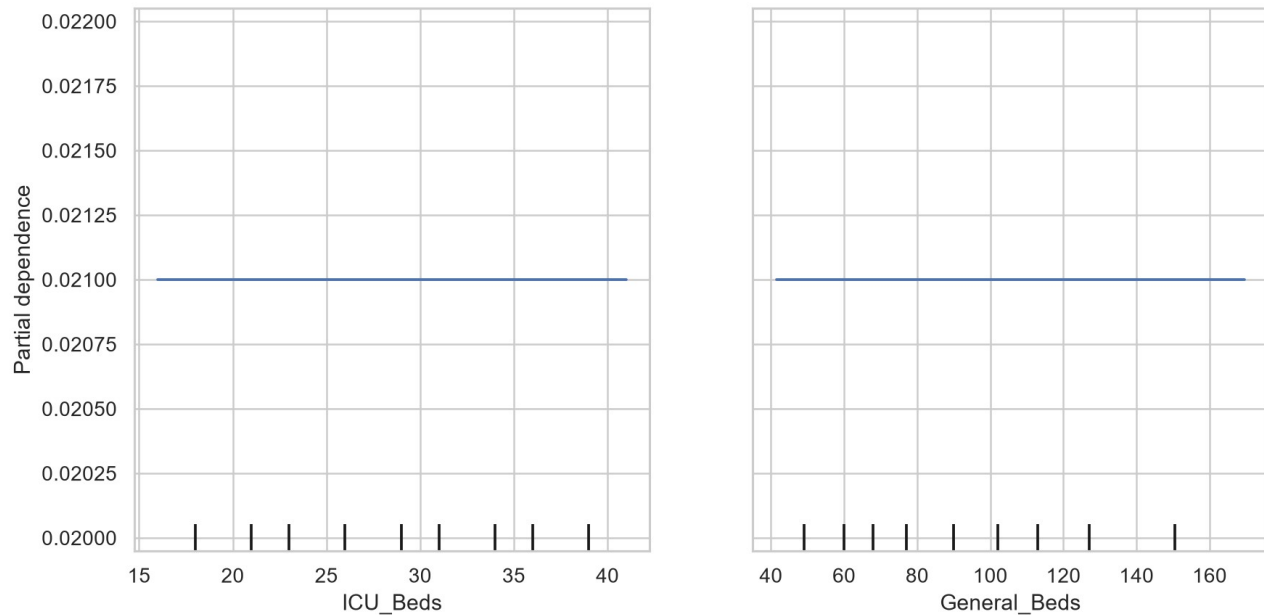


Figure 12. Partial dependence plots for selected features

15. Tkinter GUI and User Workflow

The Tkinter application provides dataset upload, preview, missing-value inspection, visualization, imputation selection, automatic recommendation, model training, prediction, export, report generation, progress feedback, logging and status handling. The package should be launched from the project root with `python -m gui.app`. A direct-script launcher can also be supported through a dedicated `run_gui.py` entry point.

- Upload or open the generated missing dataset.
- Inspect row count, columns and missingness.
- Run automatic imputation benchmarking or choose a method manually.
- Export the repaired and feature-engineered datasets.
- Train or load the saved model pipeline.
- Generate predictions and export outputs.
- Open visualizations and create the PDF report.

16. Reproducibility and Execution

```
# Create and activate a Python 3.12 environment
py -3.12 -m venv .venv
.venv\Scripts\Activate.ps1
python -m pip install --upgrade pip
pip install -r requirements.txt
```



```
# Quick pipeline  
python run_project.py --quick
```

```
# Full pipeline  
python run_project.py
```

```
# GUI  
python -m gui.app
```

17. Business and Healthcare Insights

- Feature-specific repair is preferable to a single global rule because variables exhibit different temporal behavior, scale and missingness mechanisms.
- Available bed capacity is strongly and inversely related to occupancy rate, making it a central operational capacity indicator.
- Department-level occupancy differences are statistically meaningful and justify department-specific planning thresholds.
- Holiday periods have a significant occupancy shift in the synthetic data and should be considered in staffing and capacity forecasts.
- MNAR missingness in staffing and high COVID-load conditions is especially risky because naive imputation can systematically understate operational pressure.
- Explainability outputs are effective for detecting leakage and should be included in every model-governance review.

18. Limitations

- The dataset is synthetic and cannot capture all clinical, administrative and regional complexities of real hospitals.
- The current classifier evaluation is compromised by target leakage and must be rerun after predictor correction.
- Missingness mechanisms are simulated from predefined rules rather than estimated from real hospital reporting processes.
- Imputation uncertainty is summarized through point estimates; multiple-imputation confidence intervals are not propagated into final model uncertainty.
- The desktop GUI is suitable for local use but requires additional access control, audit trails and deployment controls for institutional environments.
- The project is an analytics prototype and is not a clinical decision-support system.

19. Recommended Improvements

- Remove every target-derived feature before training and regenerate all evaluation artifacts.
- Adopt temporal cross-validation and hospital-group holdouts to test transfer across time and facilities.

- Add schema validation with Pandera and explicit business-rule checks for bed arithmetic.
- Add MLflow experiment tracking, Optuna tuning and model metadata/versioning.
- Add drift monitoring for input distributions, missingness and prediction probabilities.
- Add pytest unit, integration and pipeline smoke tests with GitHub Actions CI.
- Package the complete preprocessing-model pipeline and avoid relying on a separately fitted scaler at inference time.
- Validate the methodology on de-identified real operational data under institutional governance.

20. Conclusion

The project demonstrates a strong advanced-capstone architecture for auditable missing-data repair in hospital census analytics. Its most valuable contribution is the feature-level comparison of twelve imputation methods using both statistical reconstruction measures and downstream utility. The generated datasets, notebooks, modular source code, GUI, models, visualizations and reports form a credible portfolio foundation. The next mandatory step is leakage-safe model retraining; once target-derived predictors are removed and evaluation is repeated under temporal validation, the project will provide a substantially more defensible estimate of real-world occupancy-risk performance.

Appendix A - Key Generated Artifacts

Artifact	Purpose
data/raw/hospital_census_complete.csv	Complete synthetic reference dataset
data/raw/hospital_census_missing.csv	Dataset with MCAR, MAR and MNAR masks
data/processed/hospital_census_clean.csv	Feature-specific repaired dataset
data/engineered/hospital_census_engineered.csv	Model-ready engineered dataset
models/best_model.pkl	Serialized fitted model pipeline
models/scaler.pkl	Scaler deliverable
reports/metrics.xlsx	Consolidated metrics workbook
reports/imputation_metrics.csv	Feature-method benchmark results
reports/imputation_recommendations.csv	Automatic repair recommendations
outputs/predictions/holdout_predictions.csv	Holdout predictions
outputs/figures/*.png	EDA, evaluation and explainability figures

Appendix B - Repository Modules

Module	Responsibility
data_loader.py	Synthetic data generation and missingness injection
preprocessing.py	Cleaning, IQR and Isolation Forest analysis, preprocessing pipeline
imputation.py	Twelve repair methods and feature-level benchmarking
feature_engineering.py	Operational, temporal, lag, rolling and interaction features
visualization.py	EDA and trend visualizations
statistical_analysis.py	Normality, correlation, ANOVA and two-group tests
modeling.py	Candidate models, tuning and fitted pipeline export
evaluation.py	Classification metrics and diagnostic plots
explainability.py	SHAP, partial dependence and permutation importance
reporting.py	DOCX/PDF and workbook generation
gui/*	Desktop application and reusable widgets